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Course Title: Communication Engineering

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Lecture # 09

Multiplexing

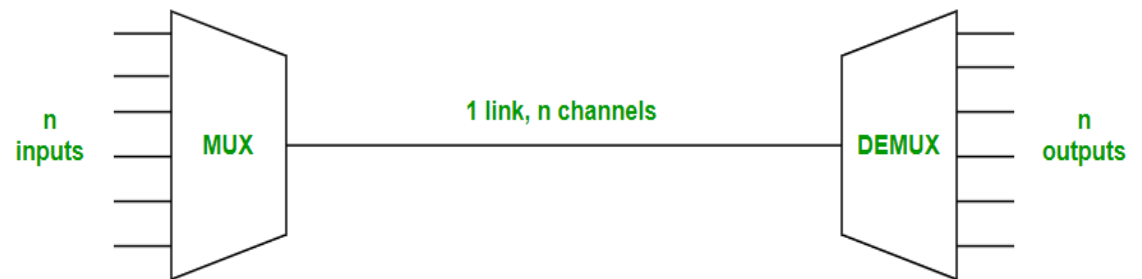
Outline_____

- **Multiplexing?**
- **Types of Multiplexing**
- **Analog: FDM, and WDM**
- **Digital: TDM, Synchronous TDM, STDM**

Multiplexing

❑ Multiplexing:

- ✓ Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a **single data link** without interference.
- ✓ The lines on the left direct their transmission streams to a **multiplexer (MUX)**, which combines them into a single stream (**many-to one**).
- ✓ At the receiving end, that stream is fed into a **demultiplexer (DEMUX)**, which separates the stream back into its component transmissions (**one-to-many**) and directs them to their corresponding lines.



Multiplexing

❑ Multiplexing:

- ✓ In the figure, the word **link** refers to the **physical path**.
- ✓ The word **channel** refers to the portion of a link that carries a transmission between a given pair of lines.
- ✓ One link can have **many (n)** channels.



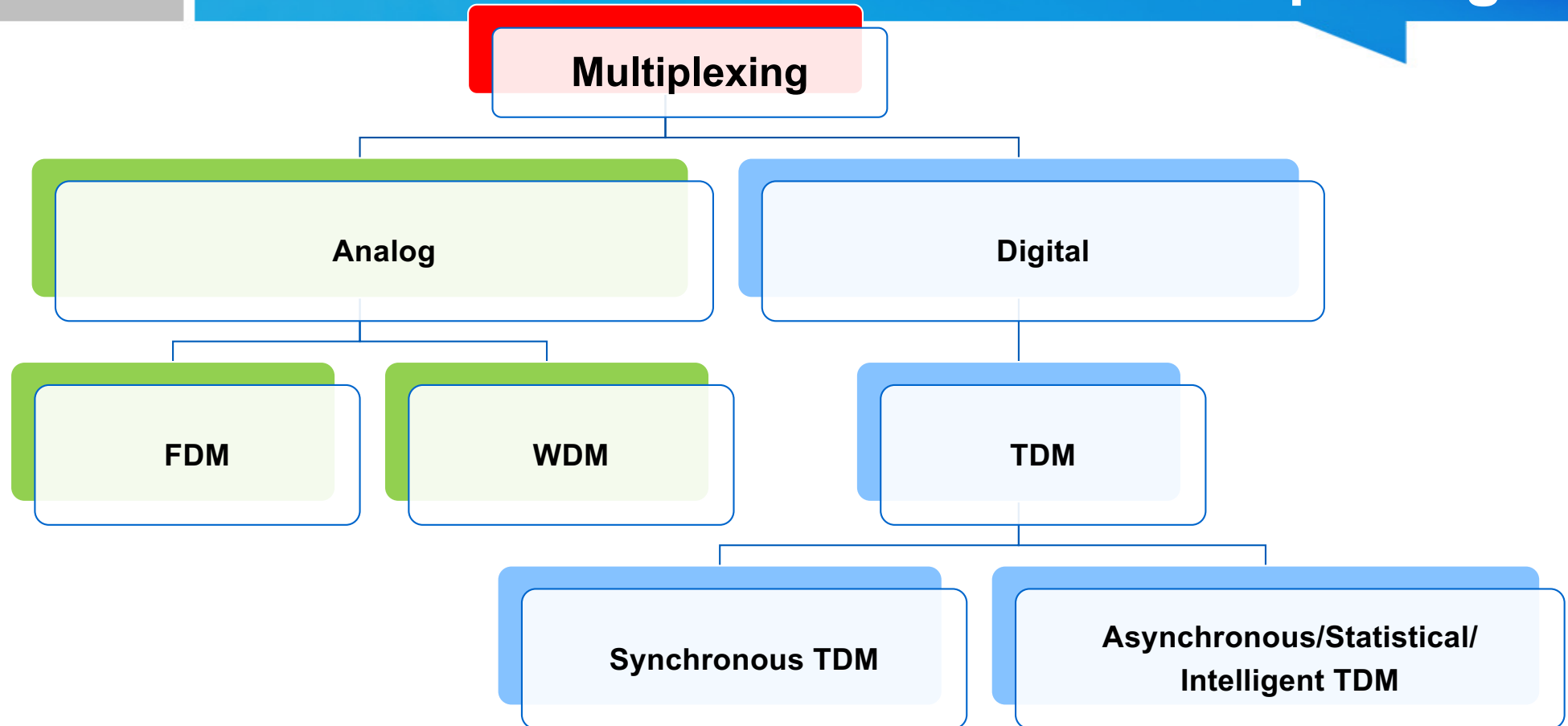
❑ Categories of Multiplexing:

❑ Signals are multiplexed by one of **three basic techniques**:

1. Frequency-division multiplexing (FDM),
2. Wave-division multiplexing (WDM), and
3. Time-division multiplexing (TDM).

✓ The first two are techniques used for analog signals; and the third for digital signals

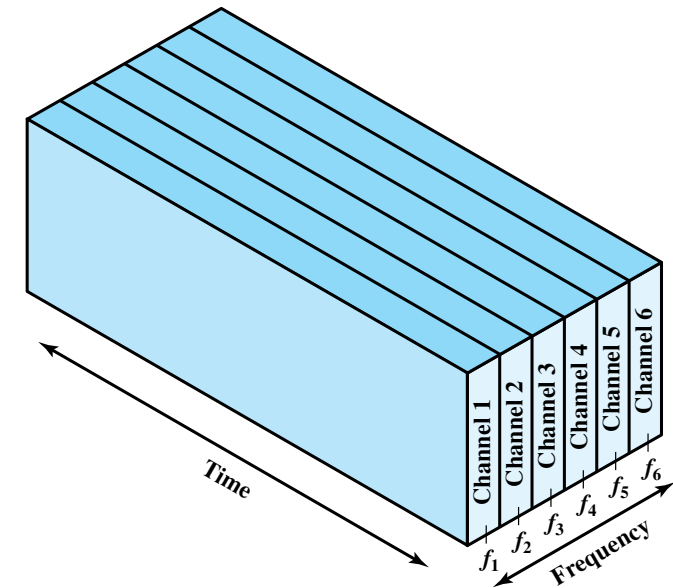
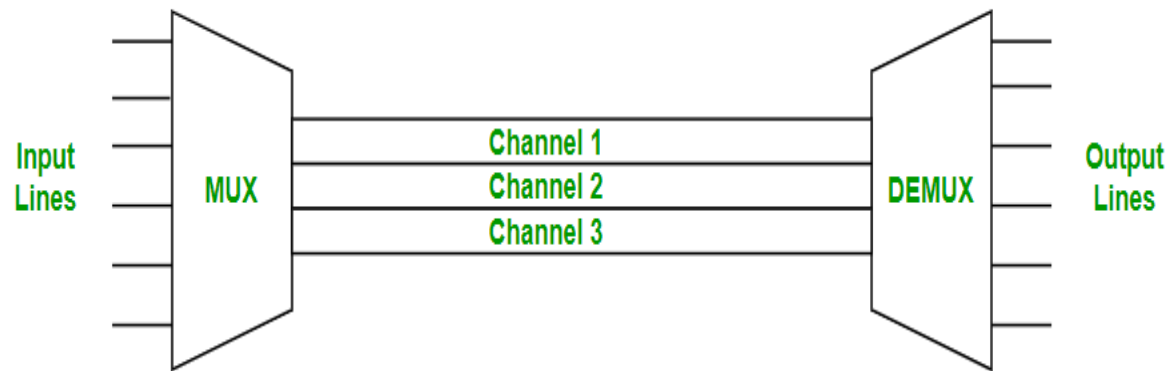
Multiplexing



Multiplexing

❑ Frequency-division multiplexing (FDM):

- ✓ FDM is defined as a type of multiplexing where the bandwidth of a single physical medium is divided into a number of smaller, independent frequency channels.



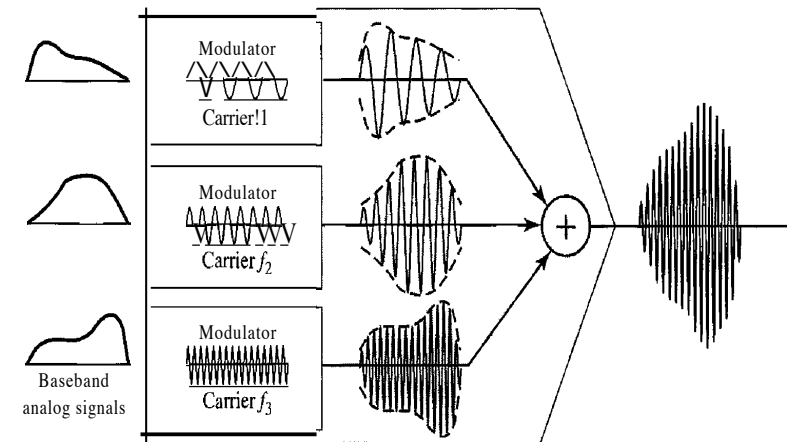
□ Frequency-division multiplexing (FDM):

- ✓ FDM can be applied when the bandwidth of a link (in hertz) is **greater** than the combined bandwidths of the signals to be transmitted.
- ✓ In FDM, signals generated by each sending device modulate **different carrier frequencies**.
- ✓ These modulated signals are then **combined into a single composite signal** that can be transported by the link.
- ✓ **Carrier frequencies** are separated by **sufficient bandwidth** to accommodate the modulated signal.
- ✓ These bandwidth ranges are **the channels** through which the various signals travel.
- ✓ Channels must be separated by strips of **unused bandwidth (guard bands)** to prevent signals from overlapping.
- ✓ In addition, carrier frequencies must not interfere with the original data frequencies.

Multiplexing

❑ Frequency-division multiplexing (FDM):

- ✓ Figure shows a conceptual illustration of the multiplexing process.
- ✓ Each source generates a signal of a similar frequency range.
- ✓ Inside the multiplexer, these similar signals modulates different carrier frequencies (f_1 , f_2 , and f_3).
- ✓ The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

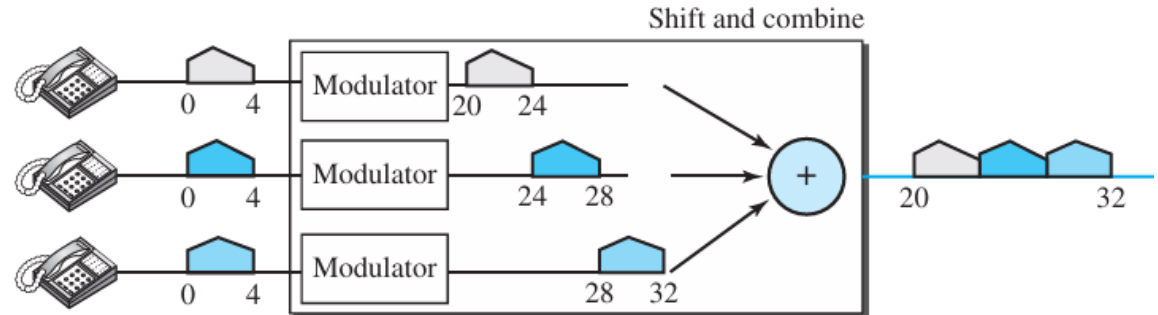


Multiplexing

❑ **Problem:** Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

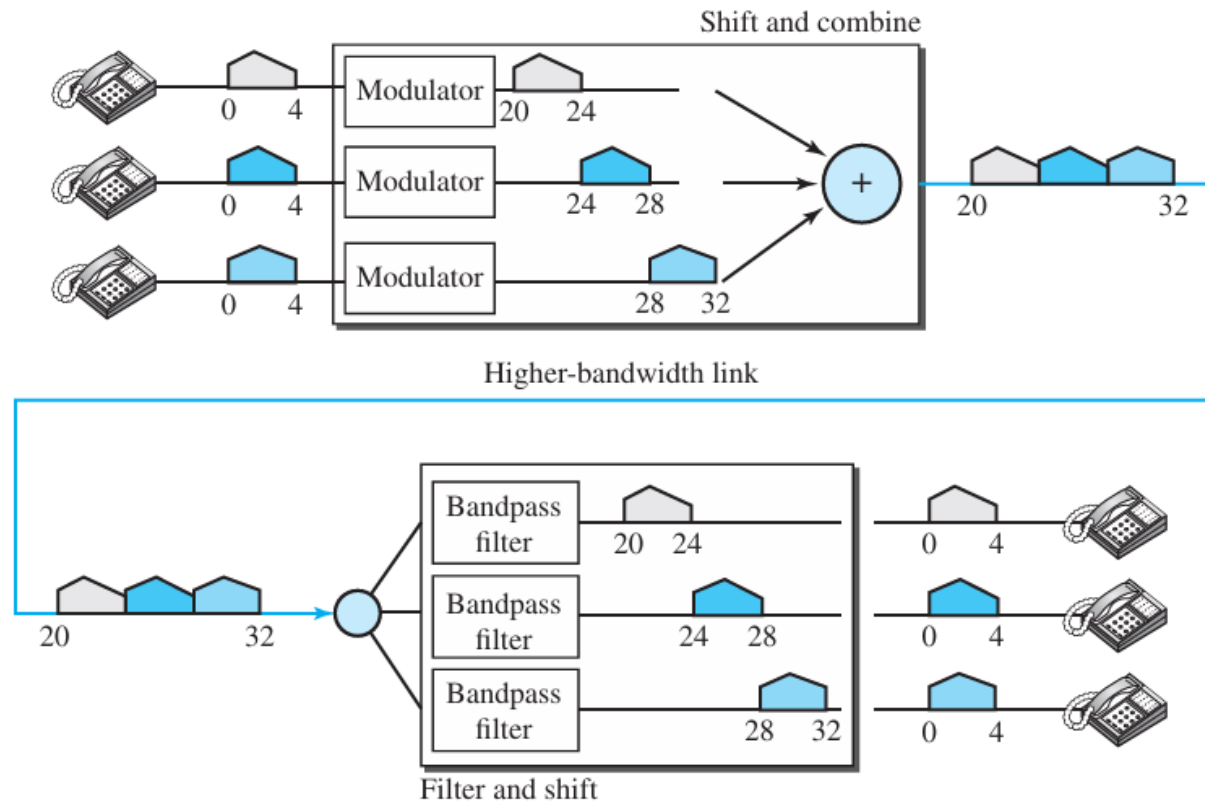
❑ **Solution:**

Shift (modulate) each of the three voice channels to a different bandwidth, as shown in figure.



Multiplexing

❑ Problem:



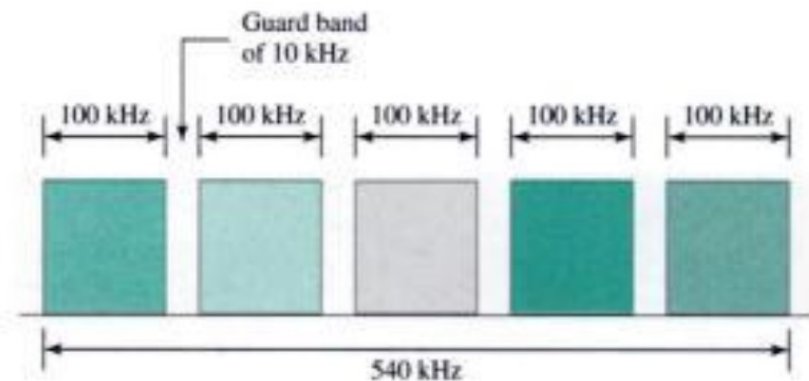
Multiplexing

❑ **Problem:** Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between the channels to prevent interference?

❑ **Solution:**

For five channels, we need at least four guard bands.

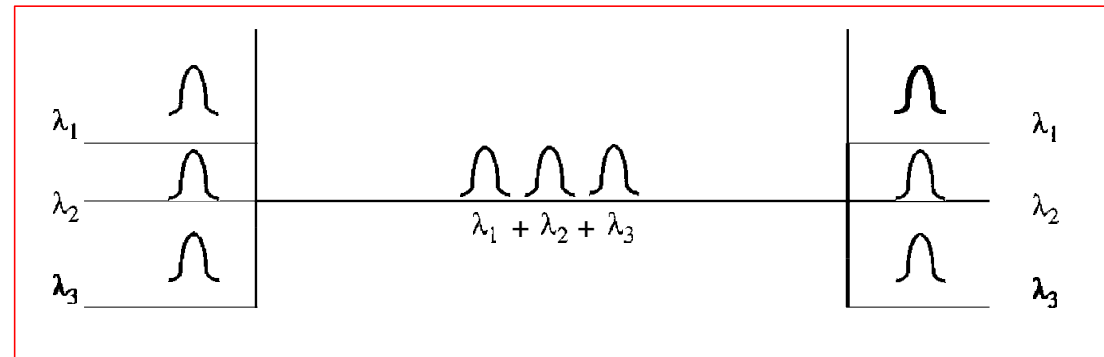
This means that the required bandwidth is at least $5 \times 100 + 4 \times 10 = 540$ KHz, as shown in the figure.



Multiplexing

❑ Wavelength-Division Multiplexing (WDM):

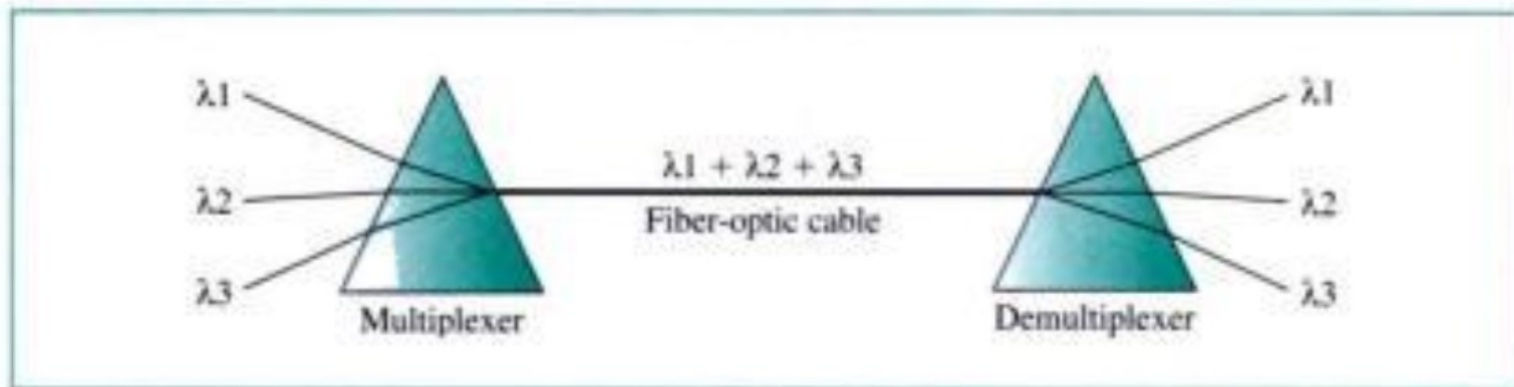
- ✓ Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable. Multiplexing allows us to combine several lines into one.
- ✓ WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- ✓ In WDM different signals of different frequencies are combined.
- ✓ The difference is that the frequencies are very high in WDM.



Multiplexing

□ Wavelength-Division Multiplexing (WDM):

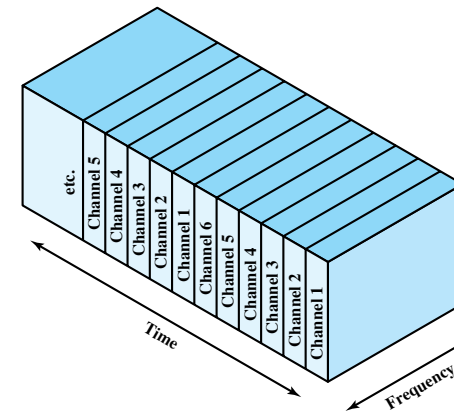
- ✓ Combine multiple light sources into one single light at the multiplexer and do the reverse at the demultiplexer. The combining and splitting of light sources are easily handled by a prism.



Multiplexing

❑ Time-Division Multiplexing (TDM):

- ✓ Time division multiplexing is defined as a type of multiplexing where time is shared. Each connection occupies a portion of time in the link.
- ✓ In Time Division Multiplexing, all signals operate with same frequency (bandwidth) at different times.
- ✓ TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

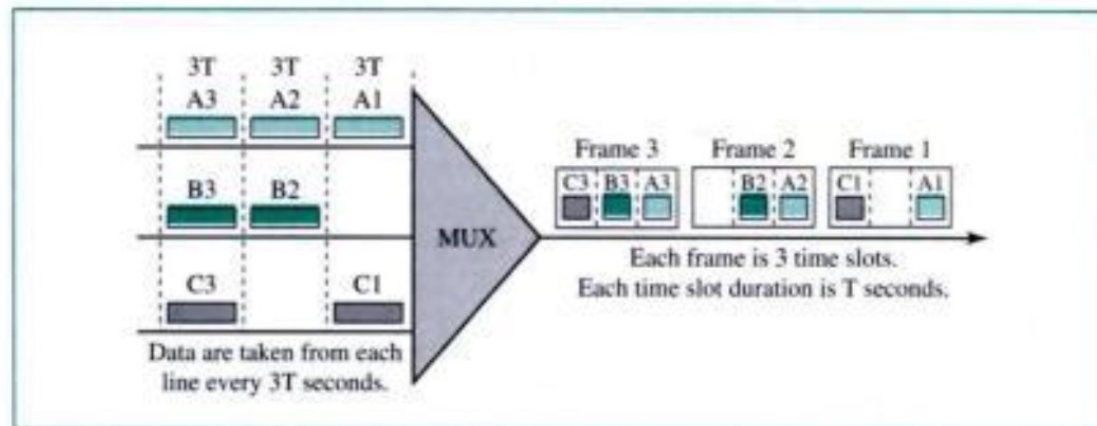


(b) Time division multiplexing

Multiplexing

❑ Synchronous Time-Division Multiplexing (STDM):

- ✓ In synchronous TDM, the data flow of each connection is divided into units, and the link combines one unit of each connection to make a frame.
- ✓ The size of the unit can be 1 bit or several bits.
- ✓ For n input connections, a frame is organized into a minimum of n time slots, each slot carrying one unit from each connection.

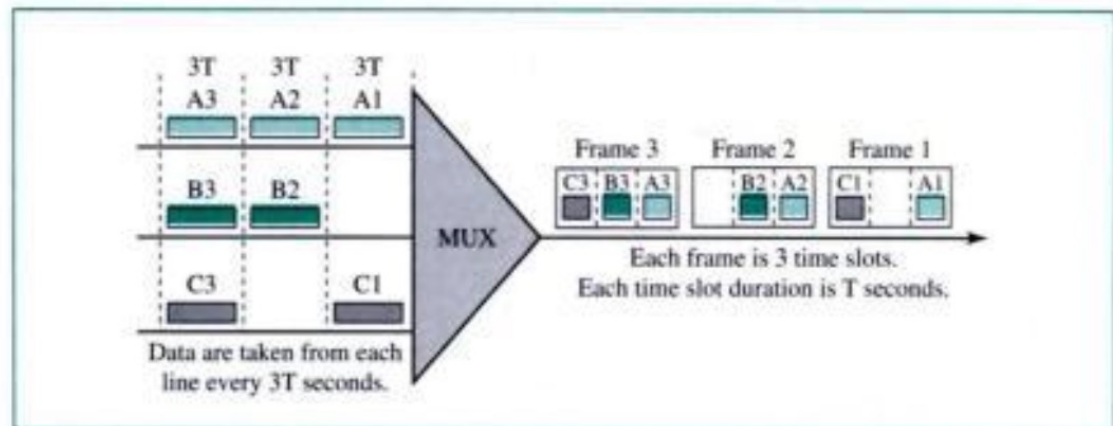


Multiplexing

❑ Synchronous Time-Division Multiplexing (STDM):

- ✓ **Empty Slots Issue:** Synchronous TDM is not as efficient as it could be. If a source does not have data to send, the corresponding slot in the output frame is empty.

- ✓ **Statistical TDM** can improve the efficiency by removing the empty slots from the frame.



Multiplexing

❑ Statistical Time-Division Multiplexing:

- ✓ In synchronous TDM, each input has a reserved slot in the output frame.
- ✓ This can be inefficient if some input lines have no data to send.
- ✓ In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.
- ✓ Only when an input line has a slot's worth of data to send is it given a slot in the output frame.
- ✓ In statistical multiplexing, the number of slots in each frame is less than the number of input lines.
- ✓ The multiplexer checks each input line in round-robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.

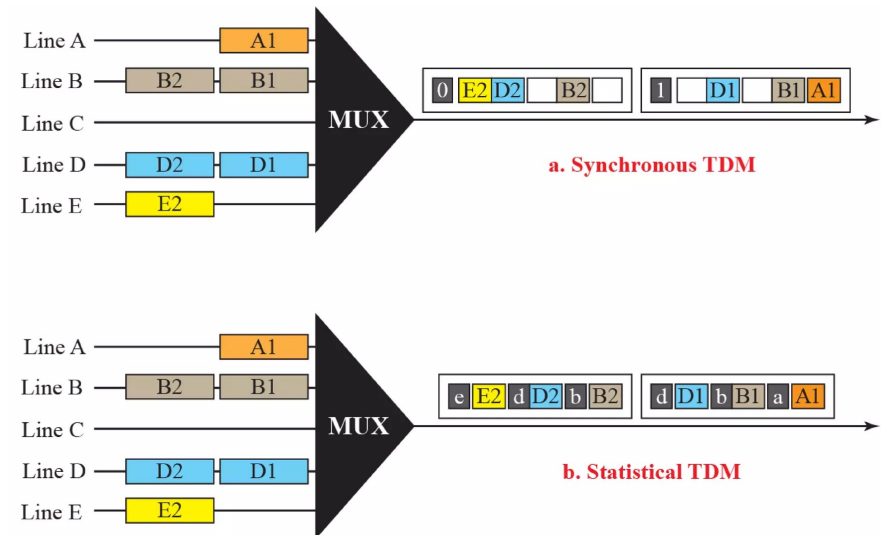


Figure: TDM slot comparison

Thank You All